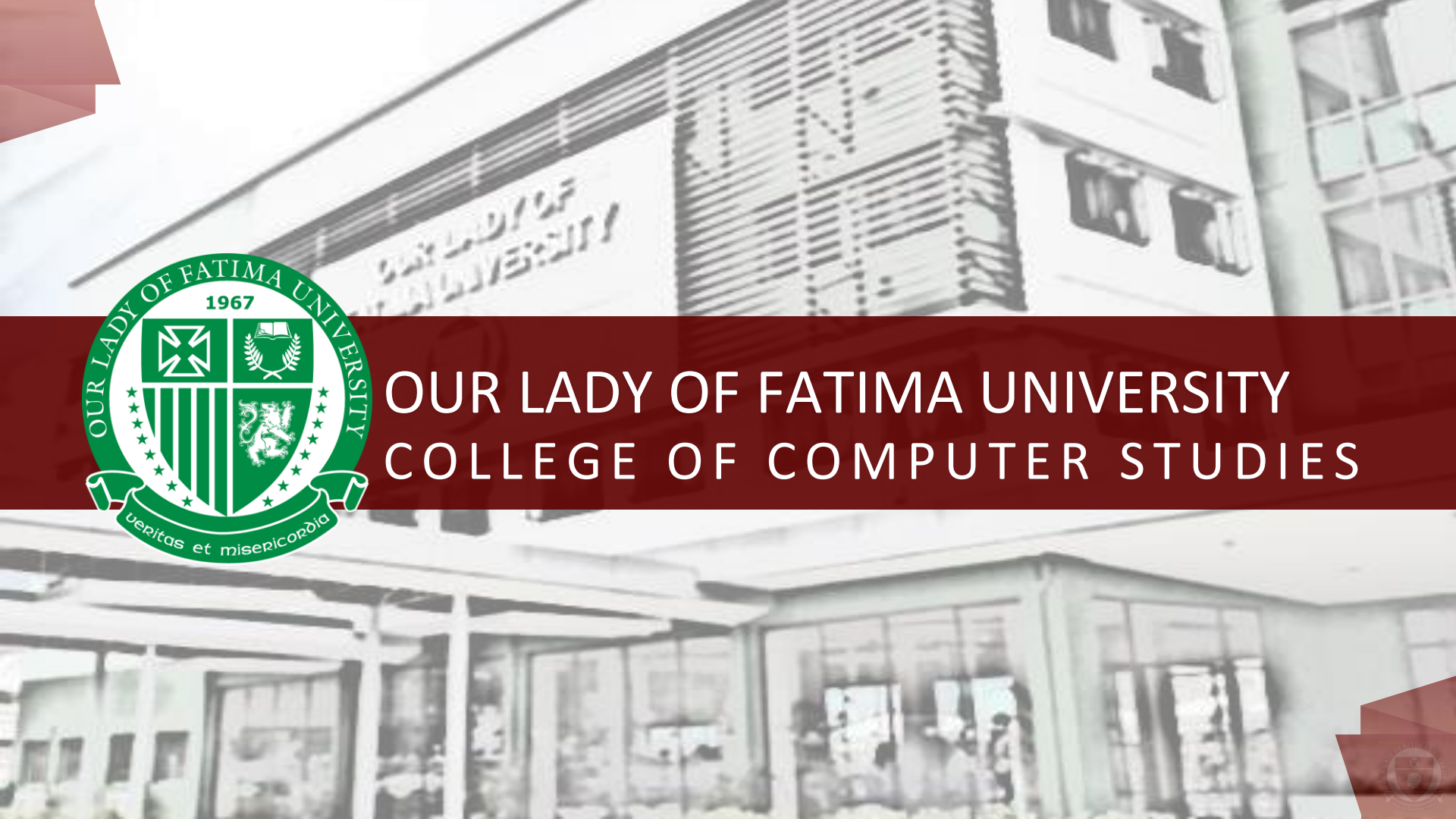
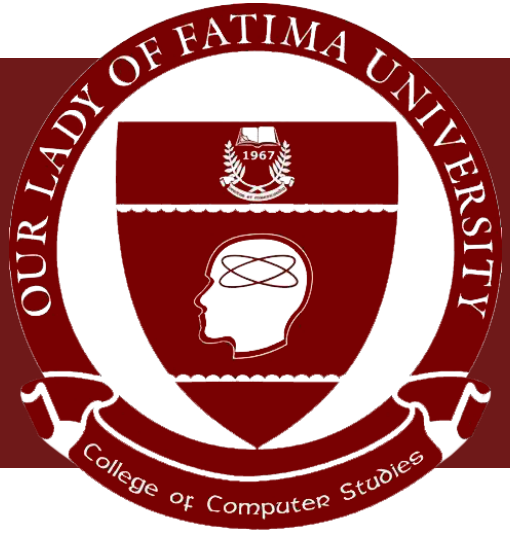
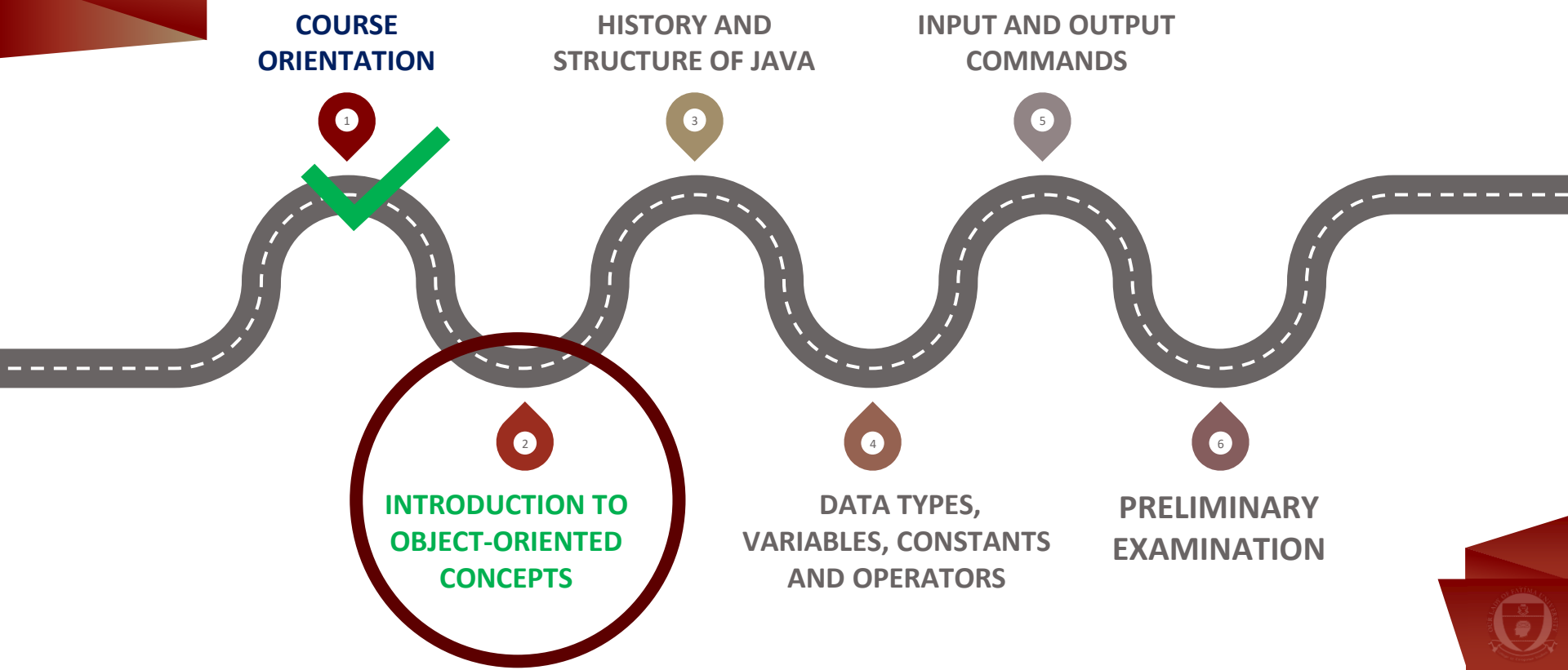


OUR LADY OF FATIMA UNIVERSITY
COLLEGE OF COMPUTER STUDIES



OOPR211
OBJECT-ORIENTED
PROGRAMMING I

PRELIMS ROADMAP



Module 1: INTRODUCTION TO OBJECT-ORIENTED CONCEPTS

Module Objectives:

1

Explain the difference between a **Procedural Programming and Object-Oriented Programming**

2

Understand **the basic concepts of Object-Oriented Programming**

Topics:

- Introduction of Programming Paradigms
- Procedural Programming vs Object Oriented Programming
- Basic Concepts in Object-Oriented Programming



1 Introduction of Programming Paradigms

Computers are so widespread in our society because they have THREE advantages over us humans.



1

Computers can store huge amounts of information.



2

Computers can recall that information quickly and accurately.

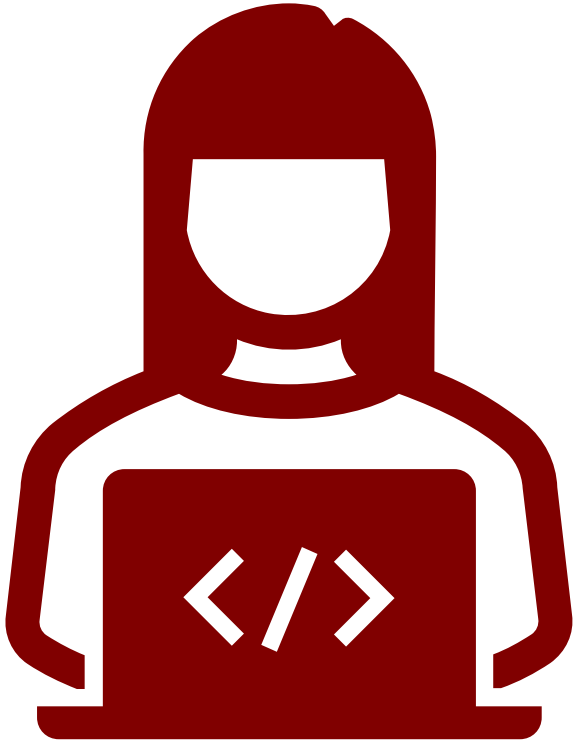


3

Computers can perform calculations with lightning speed & perfect accuracy.

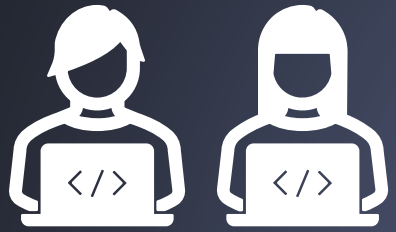
COMPUTERS…

…fundamentally are far more powerful than brain.



COMPUTERS

cannot do anything without step-by-step instructions from us telling it what to do.

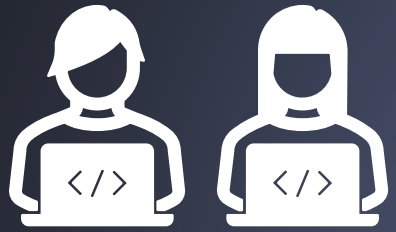


Check your Understanding

Learning about PROGRAMMING

It is known as **set of written instructions that tells computer what to do.**

a. ALGORTHIM
b. COMPUTER PROGRAM



Check your Understanding

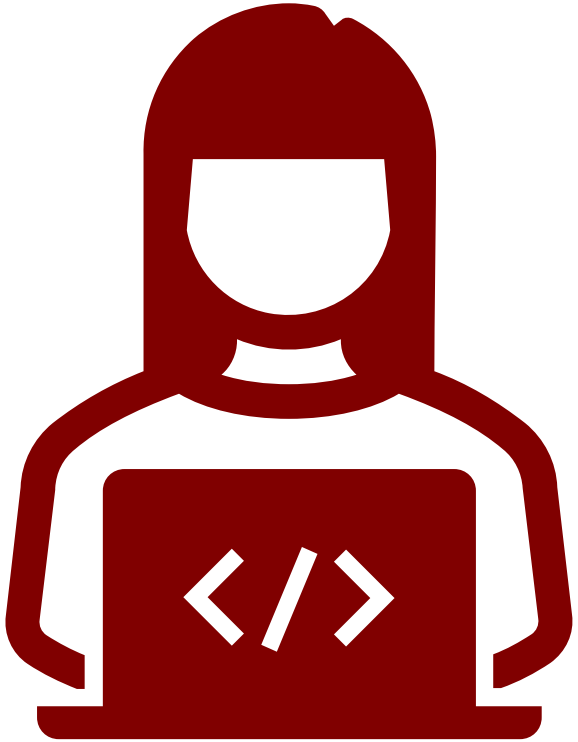
Learning about PROGRAMMING

It is known as **set of written instructions that tells computer what to do.**

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b. COMPUTER PROGRAM

COMPUTER PROGRAM

It is known as set of written instructions that tells computer what to do.



Programming Paradigms

Programming is a creative process carried out by programmers to instruct a computer on how to do a task. There are a number of alternative approaches to the programming process, referred to as programming paradigms.

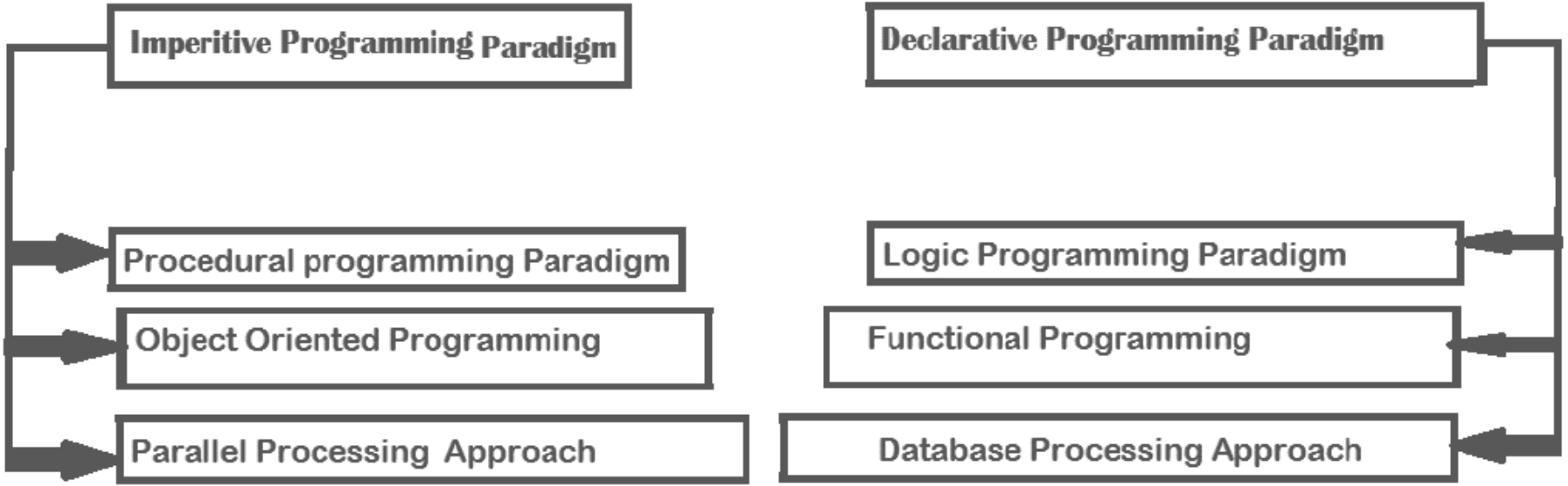
Programming Paradigms

Programming is the process of taking an algorithm and encoding it into a notation, a programming language, so that it can be executed by a computer.

Paradigms

Paradigm can also be termed as method to solve some problem or do some task. Programming paradigm is an approach to solve problem using some programming language or also we can say it is a method to solve a problem using tools and techniques that are available to us following some approach. There are lots for programming language that are known but all of them need to follow some strategy when they are implemented, and this methodology/strategy is paradigms.

Programming Paradigms



2 Popular approaches in WRITING computer programs

2

Procedural Programming vs Object Oriented Programming

OBJECT-ORIENTED VS PROCEDURAL

What is

PROCEDURAL PROGRAMMING?

PROCEDURAL PROGRAMMING

Also known as

imperative programming.

PROCEDURAL PROGRAMMING

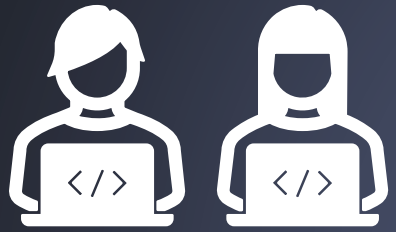
uses a list of instructions to tell the computer what to do step-by-step

PROCEDURAL PROGRAMMING

It relies on procedures or subroutines to perform computations.

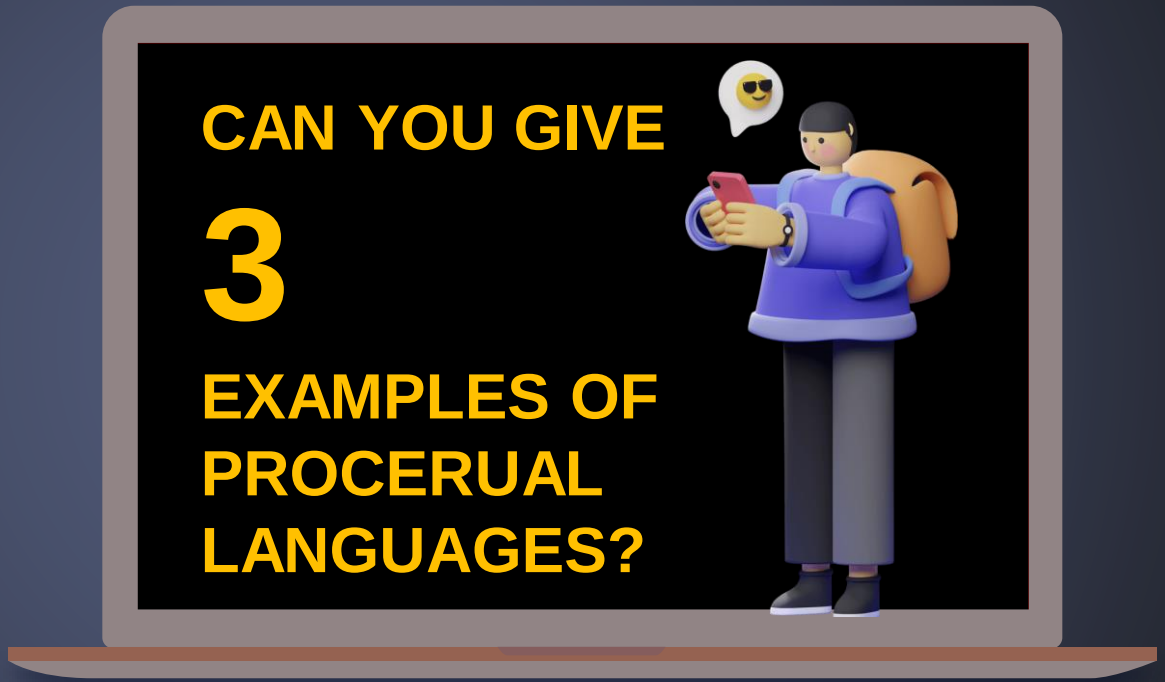


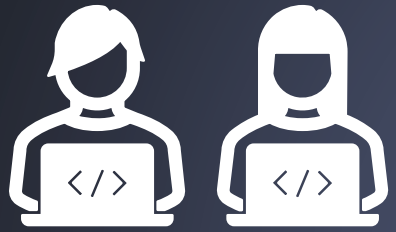
Most of the early programming languages are all PROCEDURAL



Check your Understanding

Learning about PROGRAMMING





Check your Understanding

Learning about PROGRAMMING



Examples of Procedural programming paradigm

- C : developed by Dennis Ritchie and Ken Thompson
- C++ : developed by Bjarne Stroustrup
- Java : developed by James Gosling at Sun Microsystems
- ColdFusion : developed by J J Allaire
- Pascal : developed by Niklaus Wirth

Examples of Procedural programming paradigm

```
#include <iostream>
using namespace std;
int main()
{
    int i, fact = 1, num;
    cout << "Enter any Number: ";
    cin >> number;
    for (i = 1; i <= num; i++) {
        fact = fact * i;
    }
    cout << "Factorial of " << num << " is: " << fact << endl;
    return 0;
}
```

What is

OBJECT-ORIENTED PROGRAMMING?

OBJECT-ORIENTED PROGRAMMING

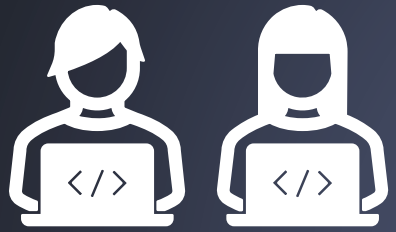
An extension of procedural programming in which you take a slightly different approach to writing computer programs.



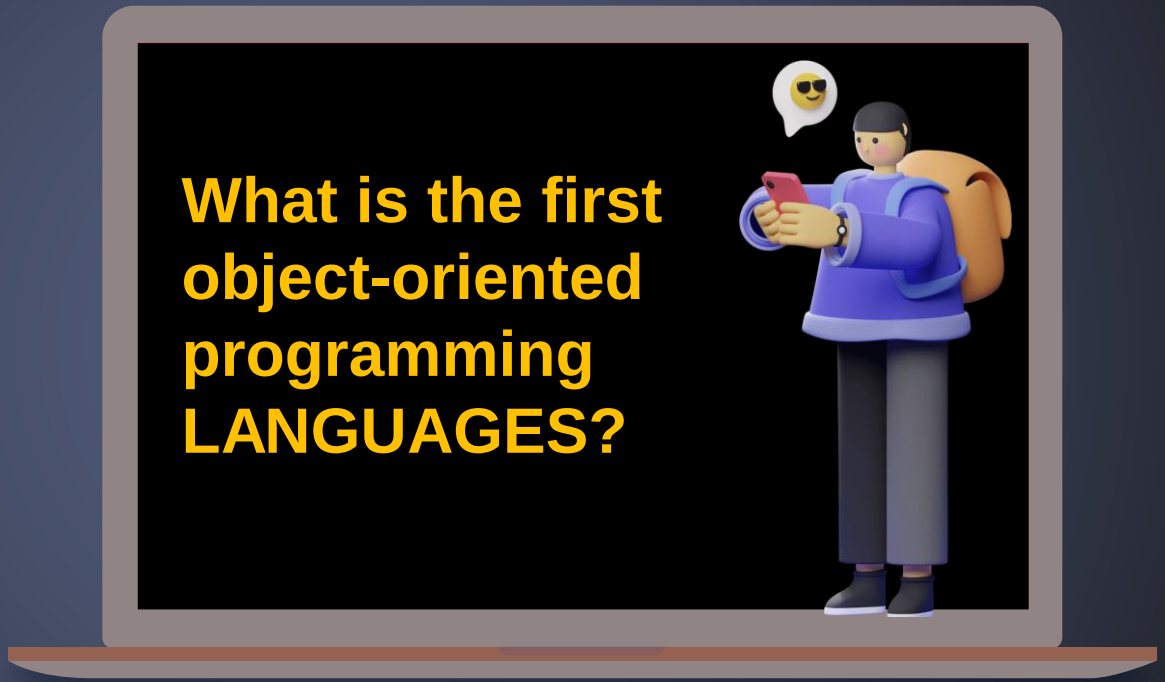
OBJECT-ORIENTED PROGRAMMING

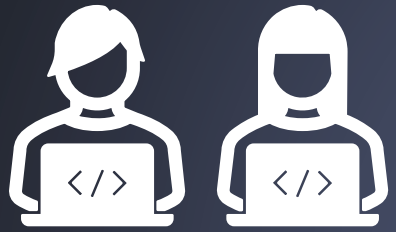
Writing object-oriented programs involves creating classes, creating objects from those classes, and creating applications.





Check your Understanding





Check your Understanding





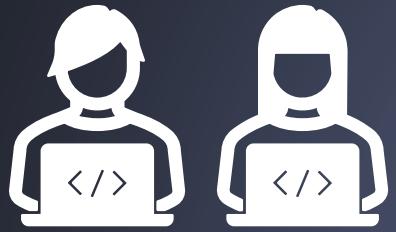
Paradigm	Object-oriented
Designed by	Ole-Johan Dahl
Developer	Kristen Nygaard
First appeared	1962; 57 years ago
Stable release	Simula 67, Simula I
Typing discipline	Static, nominative
Implementation language	ALGOL 60 (primarily; some components Simscript)
OS	Unix-like, Windows
Website	http://www.simula67.info/
Influenced by	
ALGOL 60, Simscript	
Influenced	
Object-oriented programming languages	

SIMULA IS THE NAME OF TWO SIMULATION PROGRAMMING LANGUAGES, SIMULA I AND SIMULA 67, DEVELOPED IN THE 1960S AT THE NORWEGIAN COMPUTING CENTER IN OSLO, BY OLE-JOHAN DAHL AND KRISTEN NYGAARD.

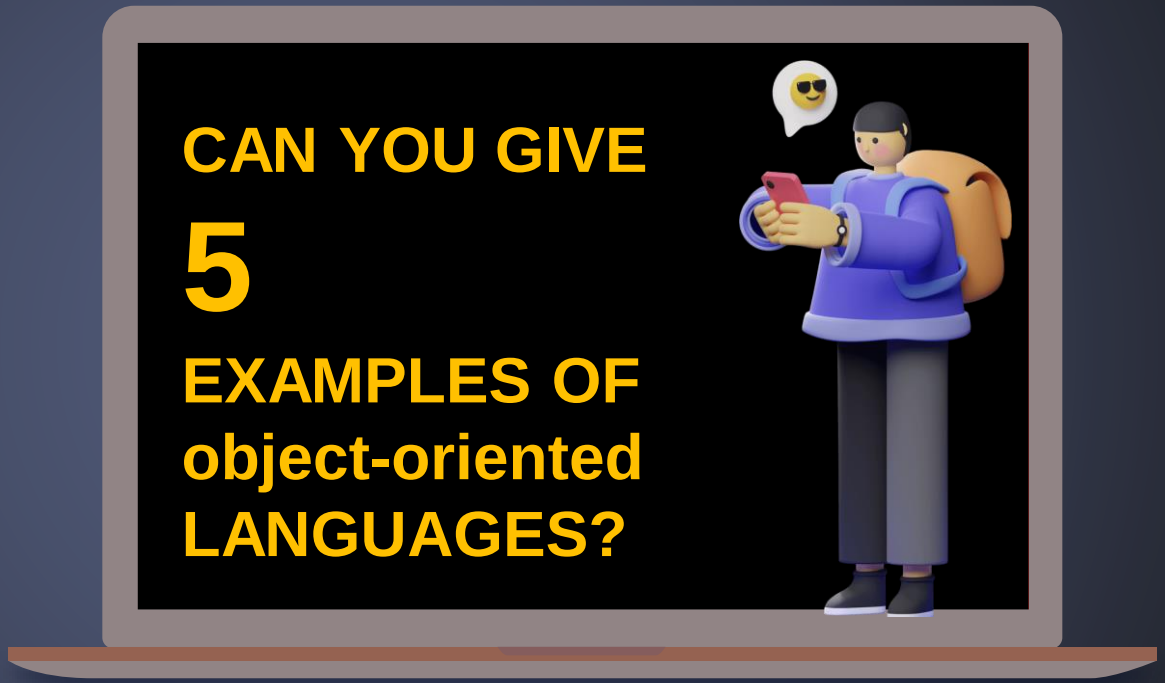
SIMULA – SAMPLE CODE

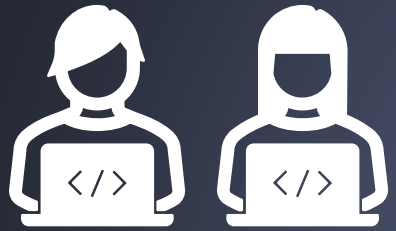
```
Begin
    OutText ("Hello, World!");
    Outimage;
End;
```

```
Glyph Class Char (c);
    Character c;
Begin
    Procedure print;
      OutChar(c);
End;
```

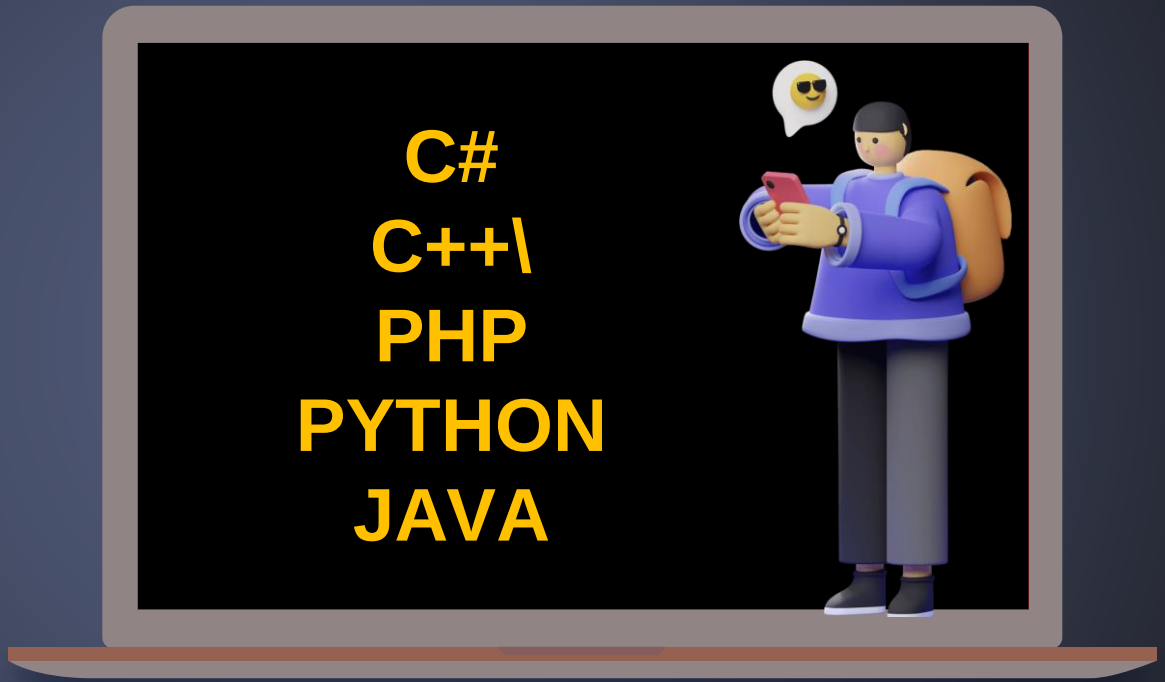


Check your Understanding





Check your Understanding



Examples of Object Oriented programming paradigm

- Simula : first OOP language
- Java : developed by James Gosling at Sun Microsystems
- C++ : developed by Bjarne Stroustrup
- Objective-C : designed by Brad Cox
- Visual Basic .NET : developed by Microsoft
- Python : developed by Guido van Rossum
- Ruby : developed by Yukihiro Matsumoto
- Smalltalk : developed by Alan Kay, Dan Ingalls, Adele Goldberg

```
import java.io.*;


class JDB {
	public static void main(String[] args)
	{
		System.out.println("WELCOME");
		Signup s1 = new Signup();
		s1.create(26, "johann", "jdbasaula@fatima.edu.ph", 'M',
4434);
	}
}
```

```
class Signup {
        int userid;  String name;  String emailid;  char sex;  long mob;
        public void create(int userid, String name,  String emailid, char sex, long mob)
        {
                System.out.println("Welcome to
                                        OOP Tutorial\nLets create your account\n");
                this.userid = userid;
                this.name = name;
                this.emailid = emailid;
                this.sex = sex;
                this.mob = mob;
                System.out.println("your account has been created");        }
```

3

Basic Concepts in Object-Oriented Programming

UNDERSTANDING

Objects & Classes



OBJECT

A COMPONENT OF A PROGRAM THAT KNOWS HOW TO PERFORM CERTAIN ACTIONS AND HOW TO INTERACT WITH OTHER ELEMENTS OF THE PROGRAM

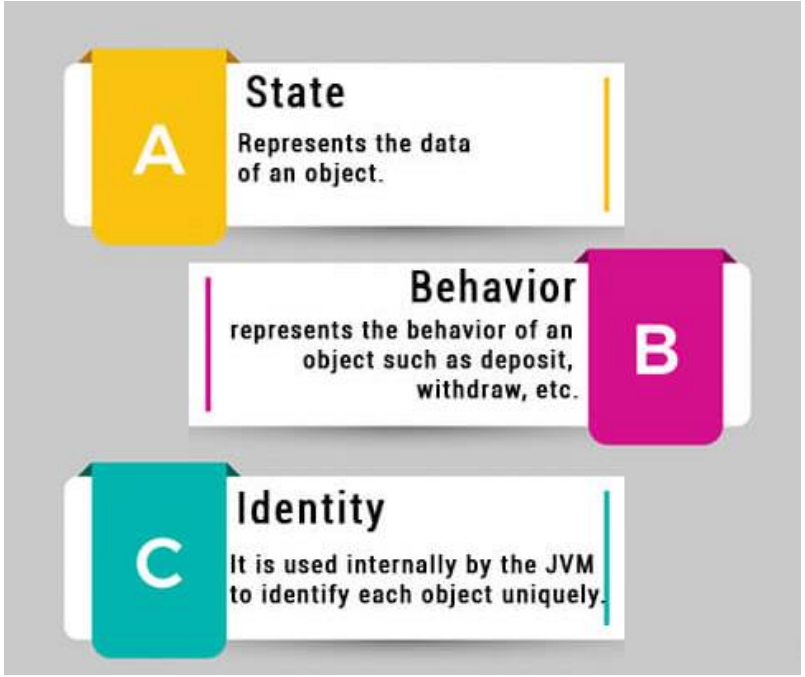
OBJECTS ARE THE BASIC UNITS OF OBJECT-ORIENTED PROGRAMMING. A SIMPLE EXAMPLE OF AN OBJECT WOULD BE A PERSON.

OBJECT

PEN IS AN OBJECT.

ITS NAME IS REYNOLDS; COLOR IS WHITE, KNOWN AS ITS STATE.

IT IS USED TO WRITE, SO WRITING IS ITS BEHAVIOR.



OBJECT

MADE UP OF ATTRIBUTES AND METHODS.

ATTRIBUTE

- CHARACTERISTICS THAT DEFINE OBJECT
- DIFFERENTIATE OBJECTS OF SAME CLASS
- VALUE OF ATTRIBUTES IS OBJECT'S STATE

METHODS

a function which is used to expose the behavior of an object.

METHODS

- Self-contained block of program code
- Similar to procedure

CLASS

A GROUP OF OBJECTS WHICH HAVE COMMON PROPERTIES.

It is a template or blueprint from which objects are created. It is a logical entity. It can't be physical.

SYNTAX TO DECLARE A CLASS

```
class <class_name>{
    field;
    method;
}
```

CORE concepts of
OBJECT-ORIENTED PROGRAMMING



INHERITANCE

WHEN ONE OBJECT ACQUIRES ALL THE PROPERTIES AND BEHAVIORS OF A PARENT OBJECT, IT IS KNOWN AS INHERITANCE.

It provides code reusability.

It is used to achieve runtime polymorphism.

INHERITANCE



JAVA CLASSES



PARENT CLASS

(SUPER OR BASE CLASS)

CHILD CLASS

(SUBCLASS OR DERIVED CLASS)

POLYMORPHISM

TAKING MANY FORMS

WHERE 'POLY' MEANS MANY AND 'MORPH' MEANS FORMS.

IT IS THE ABILITY OF A VARIABLE, FUNCTION OR OBJECT TO TAKE ON MULTIPLE FORMS.

Polymorphism allows you define one interface or method and have multiple implementations.

POLYMORPHISM

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IT IS THE ABILITY OF A VARIABLE, FUNCTION OR OBJECT TO TAKE ON MULTIPLE FORMS.

Polymorphism allows you define one interface or method and have multiple implementations.

ENCAPSULATION

A MECHANISM

WHERE YOU BIND YOUR DATA AND CODE TOGETHER

AS A SINGLE UNIT.

It also means to hide your data in order to make it safe from any modification.

ENCAPSULATION



The best way to understand encapsulation is to look at the example of a medical capsule, where the drug is always safe inside the capsule. Similarly, through encapsulation the methods and variables of a class are well hidden and safe.

ABSTRACTION

REFERS TO THE QUALITY OF DEALING WITH IDEAS RATHER THAN EVENTS.

IT BASICALLY DEALS WITH HIDING THE DETAILS AND SHOWING THE ESSENTIAL THINGS TO THE USER.

It also means to hide your data in order to make it safe from any modification.

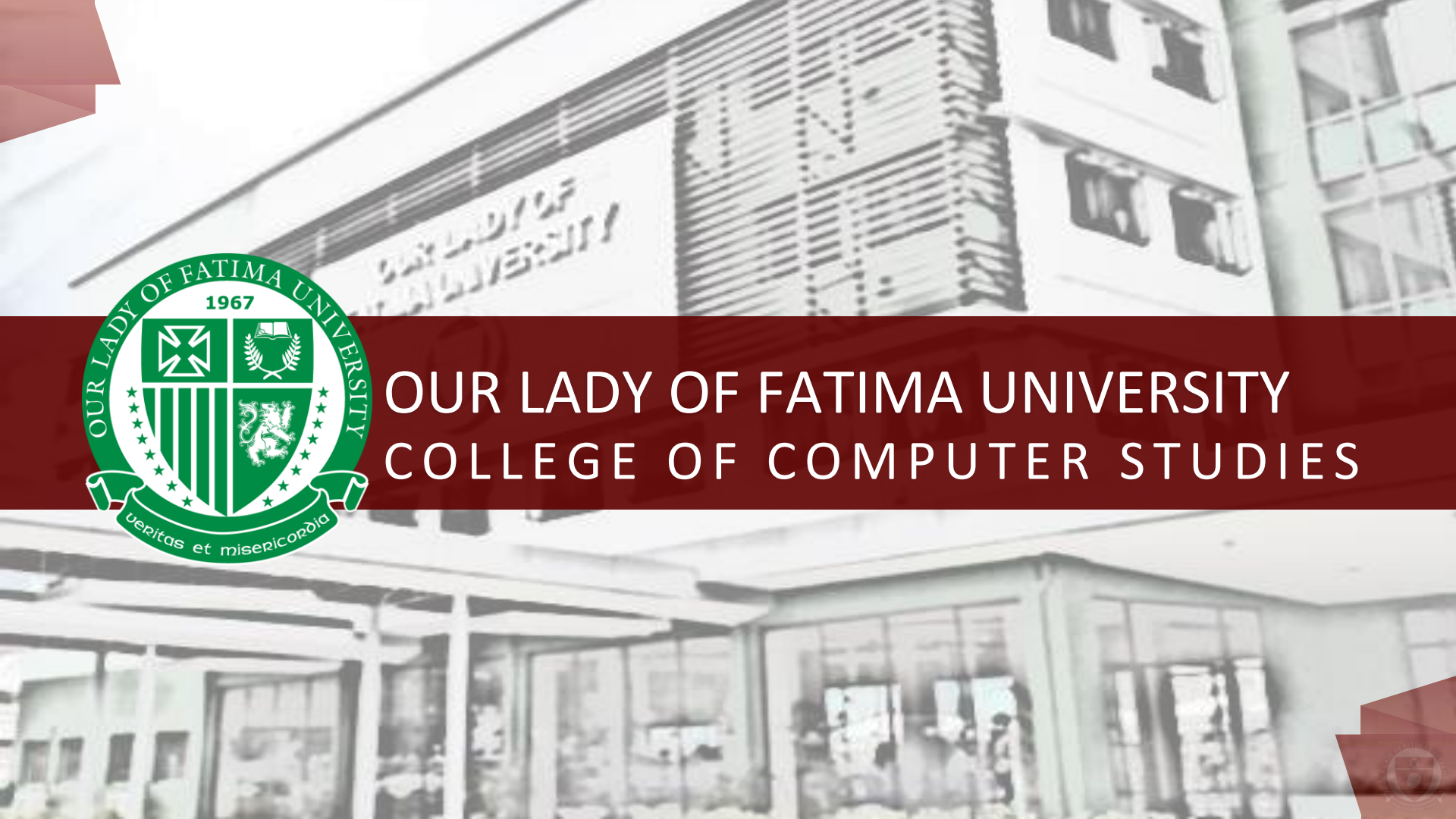
QUESTIONS?

Thank you!

Credits

Special thanks to all the people who made and released these awesome resources for free:

- Presentation template by SlidesCarnival
- Photographs by Unsplash



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